American Computer Science League

2024-2025 ● Contest 1: Short Problems ● Junior Division

1. Computer Number Systems

Convert 2024_{10} to octal.

A. 5730
B. 3814
C. 3750
D. 3754
E. 1044

2. Computer Number Systems

What is the value of $9876_{16} + ABC_{16}$ in base 16?

A. 9222
B. 9232
C. A322
D. A332
E. B332

3. Recursive Functions

Find $f(5)$ given:

$$f(x) = \begin{cases} 2 & \text{if } x \leq 1 \\ f(x-1) + x & \text{if } x > 1 \end{cases}$$

A. 11
B. 13
C. 16
D. 20
E. 22

4. Recursive Functions

Find $f(f(f(16)))$ given:

$$f(x) = \begin{cases} f(x-4) + 1 & \text{if } x \geq 10 \\ f(x+1) - 5 & \text{if } 6 < x < 10 \\ 2x & \text{if } x \leq 6 \end{cases}$$

A. 5
B. 8
C. 10
D. 12
E. 13

5. What Does This Program Do? (Branching)

What is the output of the following program?

```
x = 8
y = 9
z = -2
x = y + z
z = y - x
y = y + 1
output x, y, z
```

A. 8, 9, -2
B. 11, 10, 1
C. 7, 8, 2
D. 7, 10, -2
E. 7, 10, 2

6. What Does This Program Do? (Branching)

Run the following program with each of the following input values: 42, 35, 67, 81, 75, 24, 60, 70, 20, 16. How many of them are output?

```
input num
if num % 7 == 0 then
     output num
else
     if num % 5 == 0 then
          output num
     else
          if num >= 60 && num < 80 then
               output num
          end if
     end if
end if
```

A. 8
B. 7
C. 6
D. 5
E. 4

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2024-2025 ● Contest 1: Short Problems ● Junior Division

1. 计算机数字系统

将 2024_{10} 转换为 8 进制。

A. 5730
B. 3814
C. 3750
D. 3754
E. 1044

2. 计算机数字系统

请问 $9876_{16} + ABC_{16}$ 的值用 16 进制表示是什么？

A. 9222
B. 9232
C. A322
D. A332
E. B332

3. 递归函数

根据下列表达式，求 $f(5)$ ：

$$f(x) = \begin{cases} 2 & \text{若 } x \leq 1 \\ f(x-1) + x & \text{若 } x > 1 \end{cases}$$

A. 11
B. 13
C. 16
D. 20
E. 22

4. 递归函数

根据下列表达式，求 $f(f(f(16)))$

$$f(x) = \begin{cases} f(x-4) + 1 & \text{若 } x \geq 10 \\ f(x+1) - 5 & \text{若 } 6 < x < 10 \\ 2x & \text{若 } x \leq 6 \end{cases}$$

A. 5
B. 8
C. 10
D. 12
E. 13

5. 这个程序是做什么的？（分支）

请问**下列程序会输出什么？**

```
x = 8
y = 9
z = -2
x = y + z
z = y - x
y = y + 1
output x, y, z
```

A. 8, 9, -2
B. 11, 10, 1
C. 7, 8, 2
D. 7, 10, -2
E. 7, 10, 2

6. 这个程序是做什么的？（分支）

请**将数值** 42、35、67、81、75、24、60、70、20、16 **输入下列程序**并运行。请问有多少个值可以被输出？

```
input num
if num % 7 == 0 then
     output num
else
     if num % 5 == 0 then
          output num
     else
          if num >= 60 && num < 80 then
               output num
          end if
     end if
end if
```

A. 8
B. 7
C. 6
D. 5
E. 4